Lab 3 - Routing Protocol

NAME:

CHOH JING YI

MATRIC NUM.:

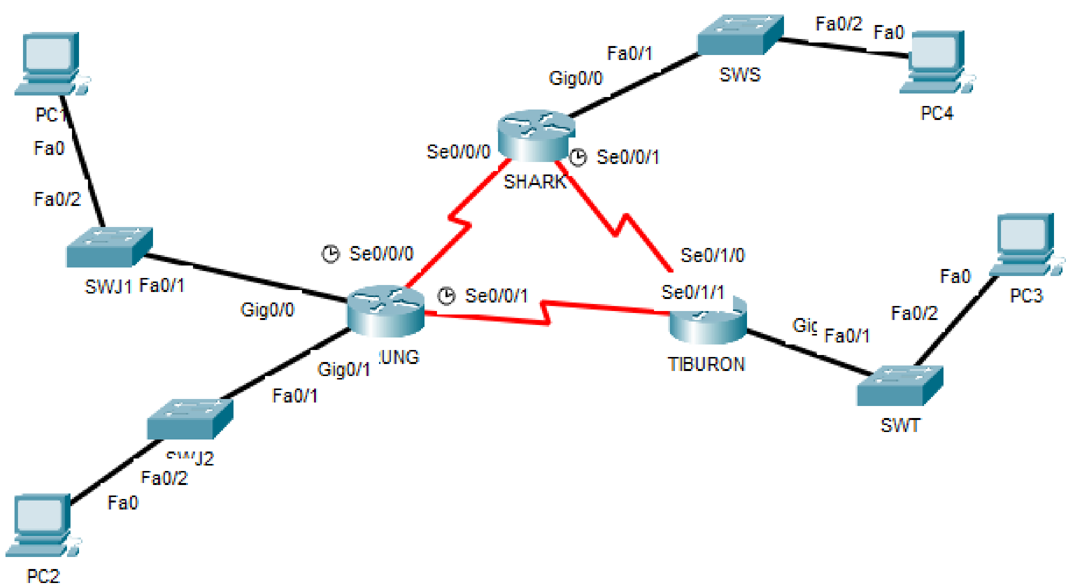
A23CS0296

SECTION:2

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	
2		Se0/0/1	172.18.110.197	255.255.255.252	
3		G0/0	172.18.110.1	255.255.255.224	
4		G0/1	172.118.110.33	255.255.255.224	
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	
6		Se0/1/0	172.18.110.202	255.255.255.252	
7		G0/0	172.118.110.129	255.255.255.192	
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	
9		Se0/0/1	172.18.110.201	255.255.255.252	
10		G0/0	172.18.110.65	255.255.255.192	
11	PC1	-	172.118.110.3	255.255.255.224	172.18.110.1
12	PC2	-	172.18.110.62	255.255.255.224	172.18.110.33
13	PC3	-	172.118.110.190	255.255.255.192	172.18.110.129
14	PC4	-	172.18.110.126	255.255.255.192	172.18.110.65

LAB TASKS

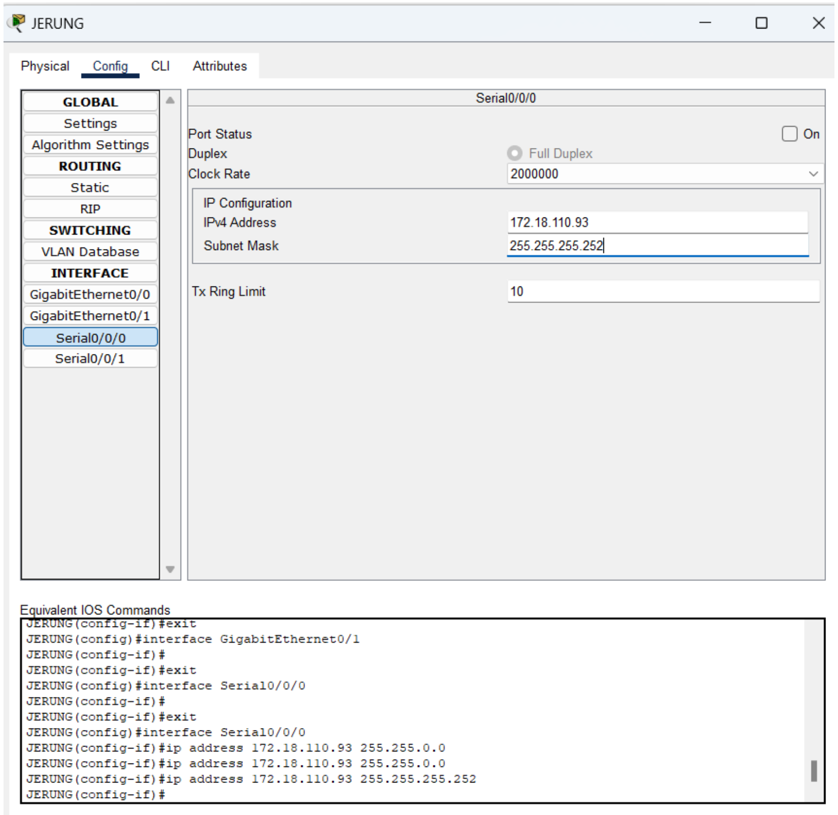
Task 1 – IP Addressing

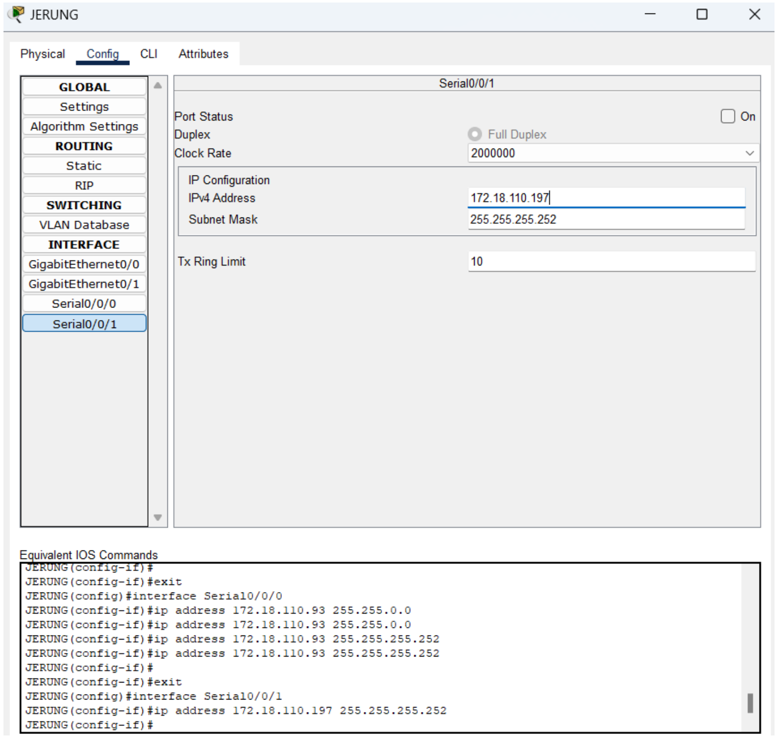
1. Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.

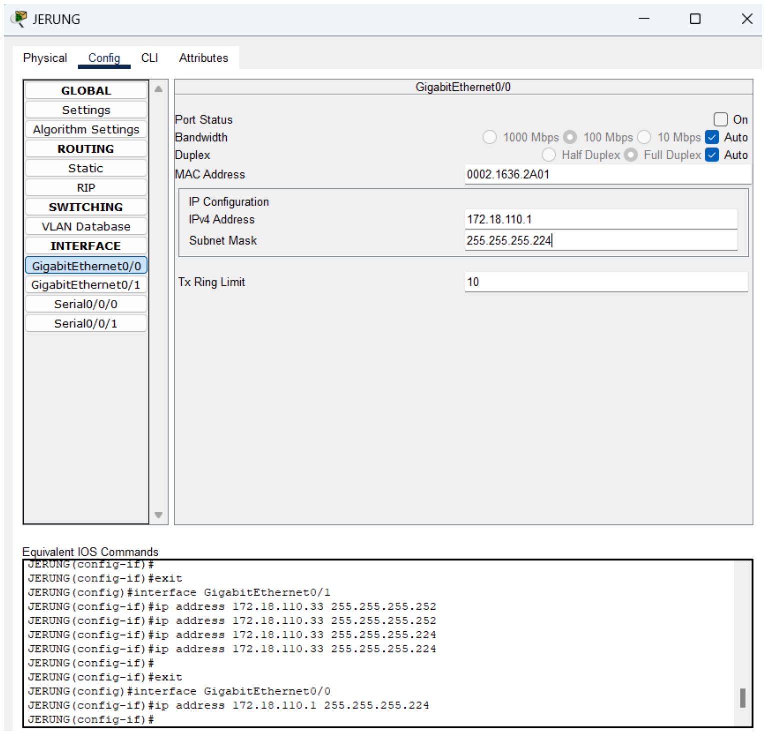
Device	Subnetwork	Usable Hosts	Network Address	Usable IP Range	Broadcast Address
JERUNG	LAN1	$20=2^5=32=/27$	172.18.110.0	172.18.110.1-172.18.110.30	172.18.110.31
	LAN2	$30=2^5=32=/27$	172.18.110.32	172.18.110.33-172.18.110.62	172.18.110.63
SHARK	LAN	$60=2^6=64=/26$	172.18.110.64	172.18.110.65-172.18.110.126	172.18.110.127
TIBURON	LAN	$50=2^6=64=/26$	172.18.110.128	172.18.110.129-172.18.110.190	172.18.110.191
Connections	JERUNG-SHARK	$2=2^2=4=/30$	172.18.110.192	172.18.110.193-172.18.110.194	172.18.110.195
	JERUNG-TRIBURON	$2=2^2=4=/30$	172.18.110.196	172.18.110.197-172.18.110.200	172.18.110.199
	SHARK-TRIBURON	$2=2^{52}=4=/30$	172.18.110.200	172.18.110.201-172.18.110.202	172.18.110.203

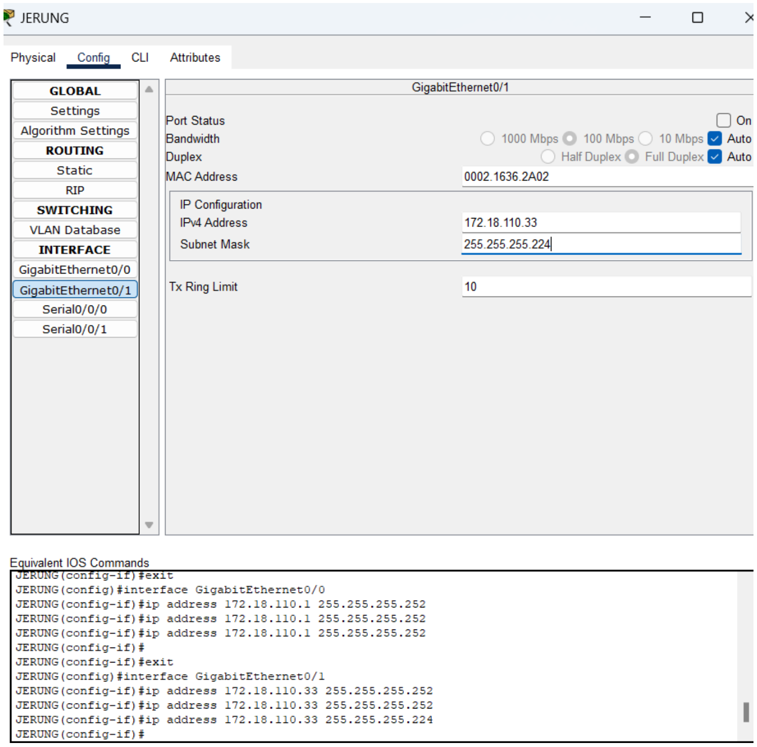
2. Using the IP addresses calculated, configure the devices with the appropriate information.

JERUNG:

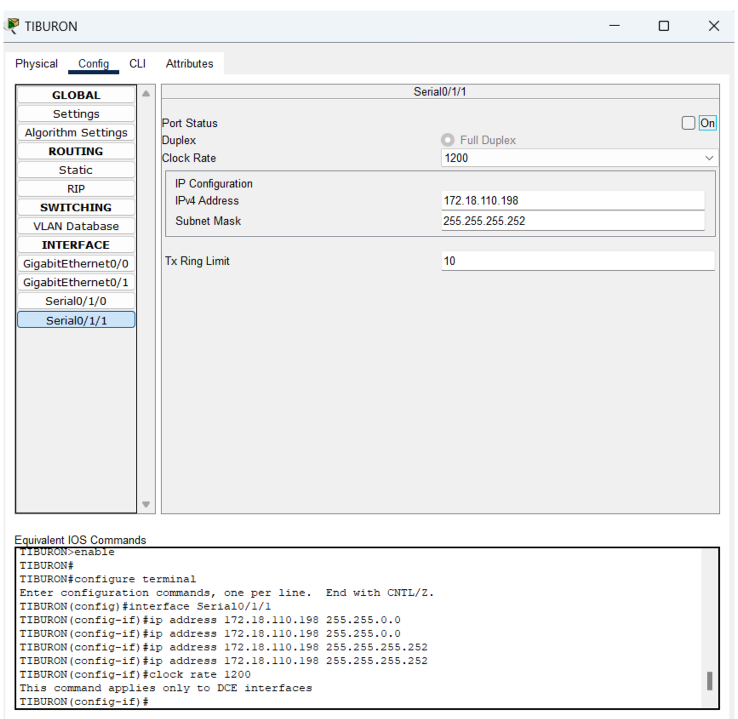


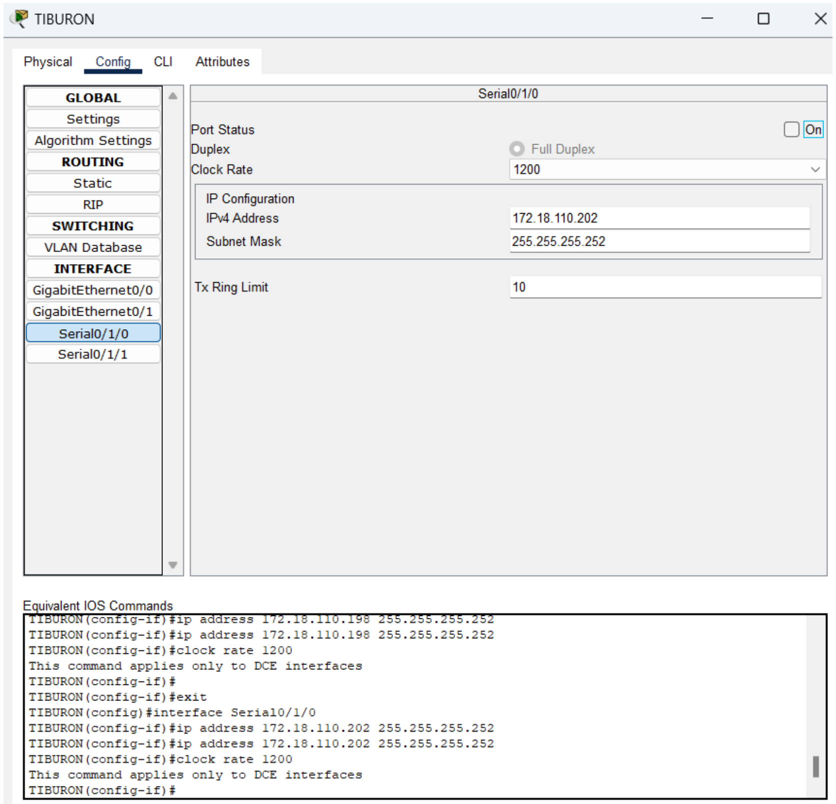

JERUNG
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/1
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
172.18.110.197
Subnet Mask
255.255.255.252
Tx Ring Limit
10
Equivalent IOS Commands
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface Serial0/0/0
JERUNG(config-if)#ip address 172.18.110.93 255.255.0.0
JERUNG(config-if)#ip address 172.18.110.93 255.255.0.0
JERUNG(config-if)#ip address 172.18.110.93 255.255.255.252
JERUNG(config-if)#ip address 172.18.110.93 255.255.255.252
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface Serial0/0/1
JERUNG(config-if)#ip address 172.18.110.197 255.255.255.252
JERUNG(config-if)#

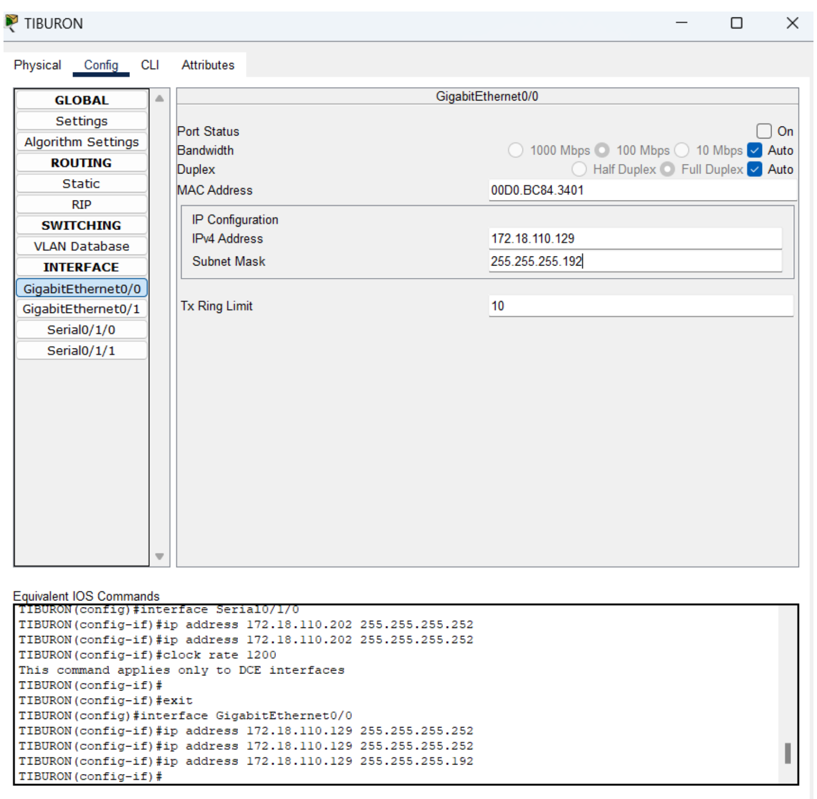

JERUNG
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
GigabitEthernet0/0
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0002.1636.2A01
IP Configuration
IPv4 Address
172.18.110.1
Subnet Mask
255.255.255.224
Tx Ring Limit
10
Equivalent IOS Commands
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface GigabitEthernet0/1
JERUNG(config-if)#ip address 172.18.110.33 255.255.255.252
JERUNG(config-if)#ip address 172.18.110.33 255.255.255.252
JERUNG(config-if)#ip address 172.18.110.33 255.255.255.224
JERUNG(config-if)#ip address 172.18.110.33 255.255.255.224
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface GigabitEthernet0/0
JERUNG(config-if)#ip address 172.18.110.1 255.255.255.224
JERUNG(config-if)#

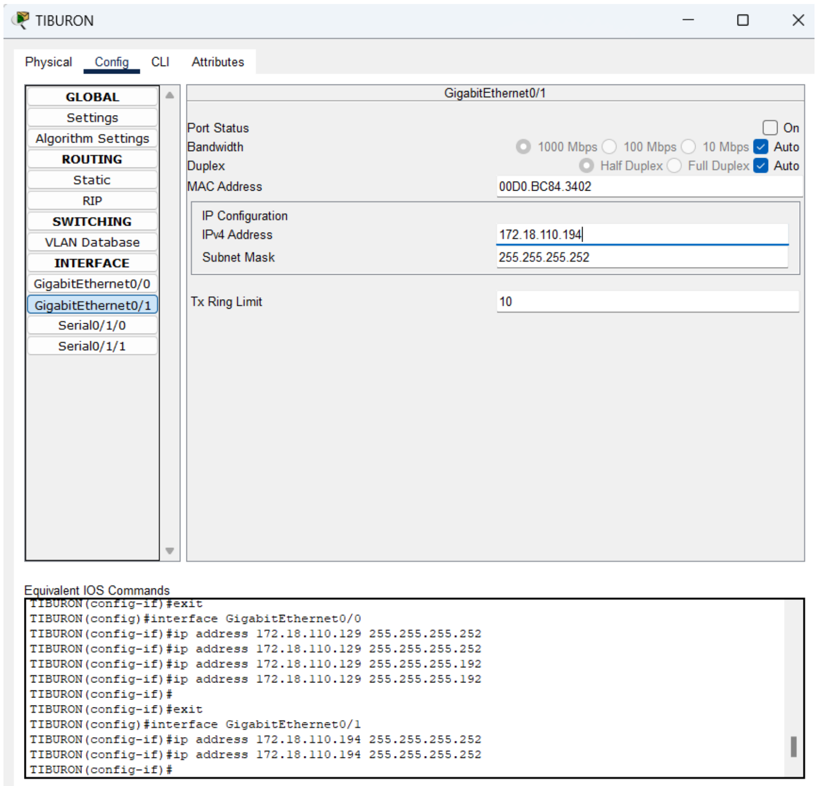

JERUNG
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
GigabitEthernet0/1
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0002.1636.2A02
IP Configuration
IPv4 Address
172.18.110.33
Subnet Mask
255.255.255.224
Tx Ring Limit
10
Equivalent IOS Commands
JERUNG(config-if)#exit
JERUNG(config)#interface GigabitEthernet0/0
JERUNG(config-if)#ip address 172.18.110.1 255.255.255.252
JERUNG(config-if)#ip address 172.18.110.1 255.255.255.252
JERUNG(config-if)#ip address 172.18.110.1 255.255.255.252
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface GigabitEthernet0/1
JERUNG(config-if)#ip address 172.18.110.33 255.255.255.252
JERUNG(config-if)#ip address 172.18.110.33 255.255.255.252
JERUNG(config-if)#ip address 172.18.110.33 255.255.255.224
JERUNG(config-if)#

TIBURON

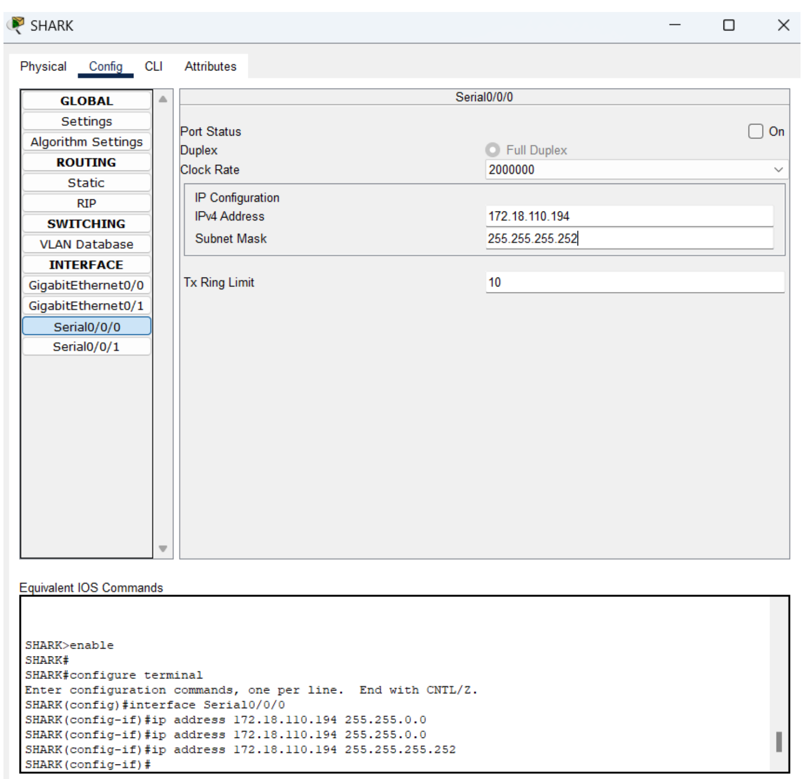

TIBURON
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/1/0
Serial0/1/1
Serial0/1/1
Port Status
On
Duplex
Full Duplex
Clock Rate
1200
IP Configuration
IPv4 Address
172.18.110.198
Subnet Mask
255.255.255.252
Tx Ring Limit
10
Equivalent IOS Commands
TIBURON>enable
TIBURON#
TIBURON#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#interface Serial0/1/1
TIBURON(config-if)#ip address 172.18.110.198 255.255.0.0
TIBURON(config-if)#ip address 172.18.110.198 255.255.0.0
TIBURON(config-if)#ip address 172.18.110.198 255.255.255.252
TIBURON(config-if)#ip address 172.18.110.198 255.255.255.252
TIBURON(config-if)#clock rate 1200
This command applies only to DCE interfaces
TIBURON(config-if)#

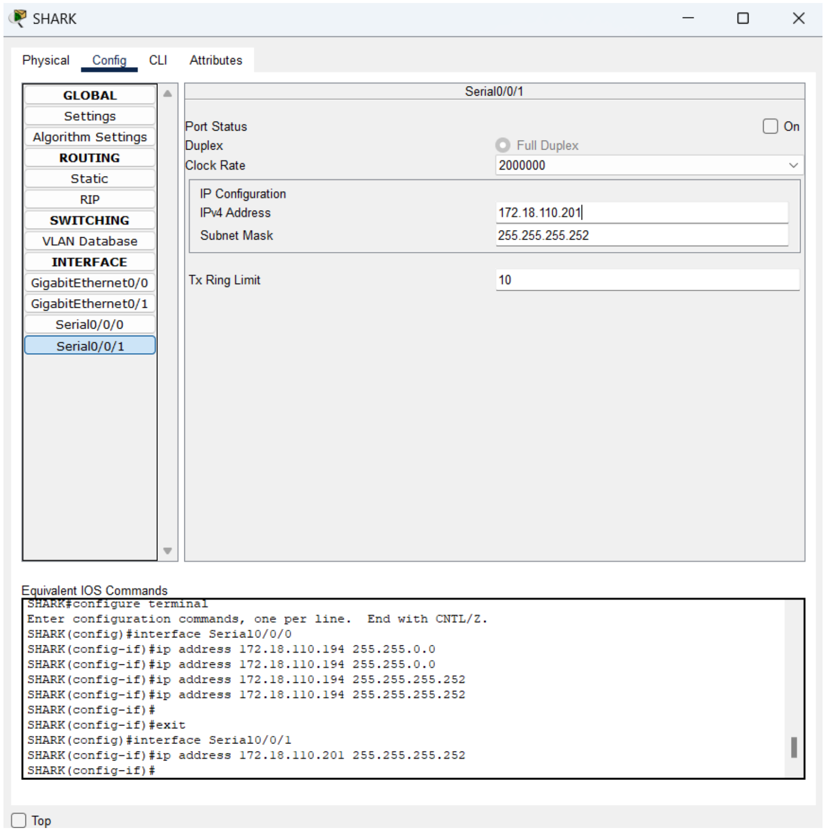

TIBURON
Physical Config CLI Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/1/0
Serial0/1/1
Serial0/1/0
Port Status
On
Duplex
Full Duplex
Clock Rate
1200
IP Configuration
IPv4 Address
172.18.110.202
Subnet Mask
255.255.255.252
Tx Ring Limit
10
Equivalent IOS Commands
TIBURON(config-if)#ip address 172.18.110.198 255.255.255.252
TIBURON(config-if)#ip address 172.18.110.198 255.255.255.252
TIBURON(config-if)#clock rate 1200
This command applies only to DCE interfaces
TIBURON(config-if)#
TIBURON(config-if)#exit
TIBURON(config)#interface Serial0/1/0
TIBURON(config-if)#ip address 172.18.110.202 255.255.255.252
TIBURON(config-if)#ip address 172.18.110.202 255.255.255.252
TIBURON(config-if)#clock rate 1200
This command applies only to DCE interfaces
TIBURON(config-if)#

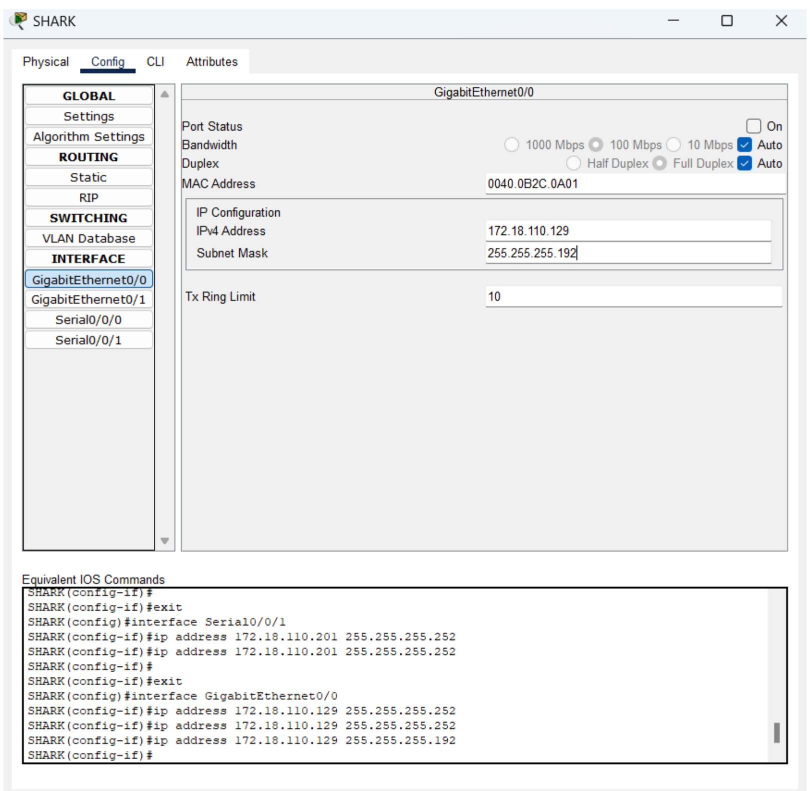

TIBURON
Physical Config CLI Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/1/0
Serial0/1/1
GigabitEthernet0/0
Port Status
On
Bandwidth
1000 Mbps 100 Mbps 10 Mbps Auto
Duplex
Half Duplex Full Duplex Auto
MAC Address
00D0.BC84.3401
IP Configuration
IPv4 Address
172.18.110.129
Subnet Mask
255.255.255.192
Tx Ring Limit
10
Equivalent IOS Commands
TIBURON(config)#interface Serial0/1/0
TIBURON(config-if)#ip address 172.18.110.202 255.255.255.252
TIBURON(config-if)#ip address 172.18.110.202 255.255.255.252
TIBURON(config-if)#clock rate 1200
This command applies only to DCE interfaces
TIBURON(config-if)#
TIBURON(config-if)#exit
TIBURON(config)#interface GigabitEthernet0/0
TIBURON(config-if)#ip address 172.18.110.129 255.255.255.252
TIBURON(config-if)#ip address 172.18.110.129 255.255.255.252
TIBURON(config-if)#ip address 172.18.110.129 255.255.255.192
TIBURON(config-if)#


TIBURON
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/1/0
Serial0/1/1
GigabitEthernet0/1
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00D0.BC84.3402
IP Configuration
IPv4 Address
172.18.110.194
Subnet Mask
255.255.255.252
Tx Ring Limit
10
Equivalent IOS Commands
TIBURON(config-if)#exit
TIBURON(config)#interface GigabitEthernet0/0
TIBURON(config-if)#ip address 172.18.110.129 255.255.255.252
TIBURON(config-if)#ip address 172.18.110.129 255.255.255.252
TIBURON(config-if)#ip address 172.18.110.129 255.255.255.192
TIBURON(config-if)#ip address 172.18.110.129 255.255.255.192
TIBURON(config-if)#
TIBURON(config-if)#exit
TIBURON(config)#interface GigabitEthernet0/1
TIBURON(config-if)#ip address 172.18.110.194 255.255.255.252
TIBURON(config-if)#ip address 172.18.110.194 255.255.255.252
TIBURON(config-if)#

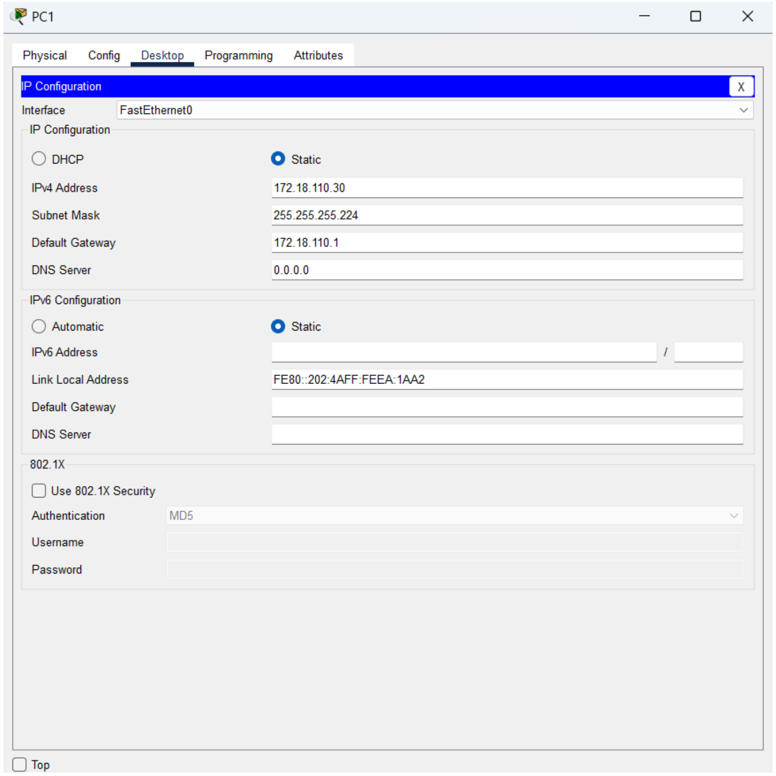
SHARK


SHARK
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
172.18.110.194
Subnet Mask
255.255.255.252
Tx Ring Limit
10
Equivalent IOS Commands
SHARK>enable
SHARK#
SHARK#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SHARK(config)#interface Serial0/0/0
SHARK(config-if)#ip address 172.18.110.194 255.255.0.0
SHARK(config-if)#ip address 172.18.110.194 255.255.0.0
SHARK(config-if)#ip address 172.18.110.194 255.255.255.252
SHARK(config-if)#

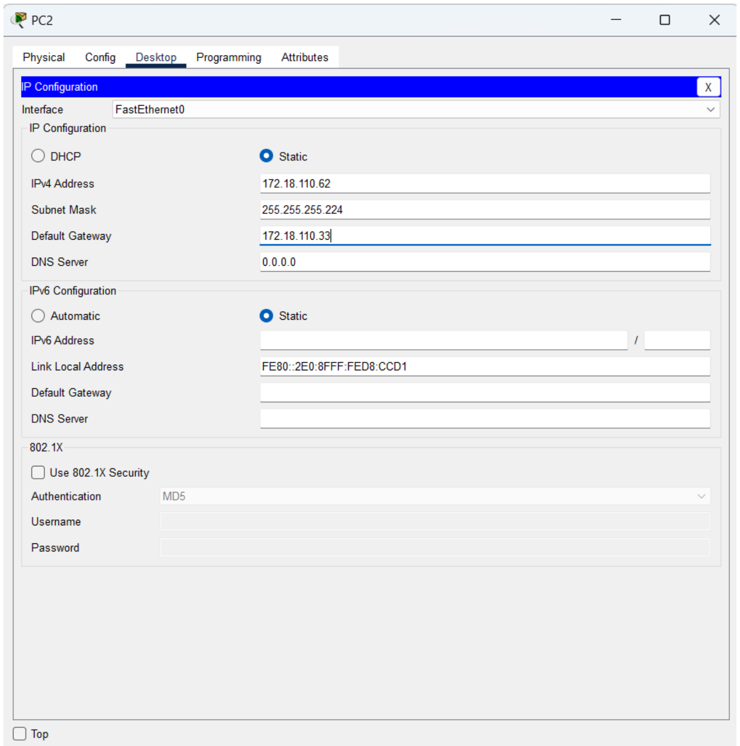

SHARK
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/1
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
172.18.110.201
Subnet Mask
255.255.255.252
Tx Ring Limit
10
Equivalent IOS Commands
SHARK#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SHARK(config)#interface Serial0/0/0
SHARK(config-if)#ip address 172.18.110.194 255.255.0.0
SHARK(config-if)#ip address 172.18.110.194 255.255.0.0
SHARK(config-if)#ip address 172.18.110.194 255.255.255.252
SHARK(config-if)#ip address 172.18.110.194 255.255.255.252
SHARK(config-if)#
SHARK(config-if)#exit
SHARK(config)#interface Serial0/0/1
SHARK(config-if)#ip address 172.18.110.201 255.255.255.252
SHARK(config-if)#
Top


SHARK
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
GigabitEthernet0/0
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0040.0B2C.0A01
IP Configuration
IPv4 Address
172.18.110.129
Subnet Mask
255.255.255.192
Tx Ring Limit
10
Equivalent IOS Commands
SHARK(config-if)#
SHARK(config-if)#exit
SHARK(config)#interface Serial0/0/1
SHARK(config-if)#ip address 172.18.110.201 255.255.255.252
SHARK(config-if)#ip address 172.18.110.201 255.255.255.252
SHARK(config-if)#
SHARK(config-if)#exit
SHARK(config)#interface GigabitEthernet0/0
SHARK(config-if)#ip address 172.18.110.129 255.255.255.252
SHARK(config-if)#ip address 172.18.110.129 255.255.255.252
SHARK(config-if)#ip address 172.18.110.129 255.255.255.192
SHARK(config-if)#

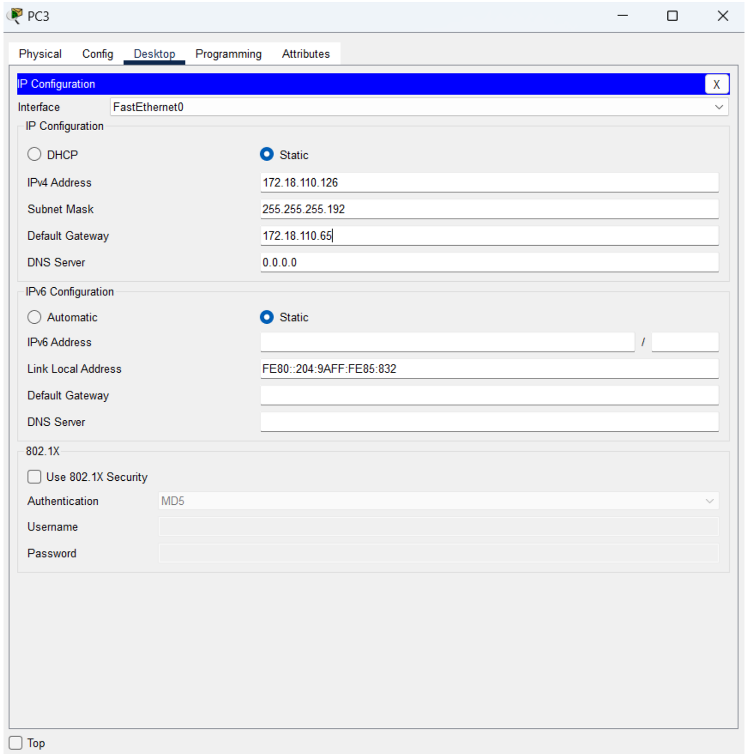
PC1


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.18.110.30
Subnet Mask
255.255.255.224
Default Gateway
172.18.110.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::202:4AFF:FEEA:1AA2
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

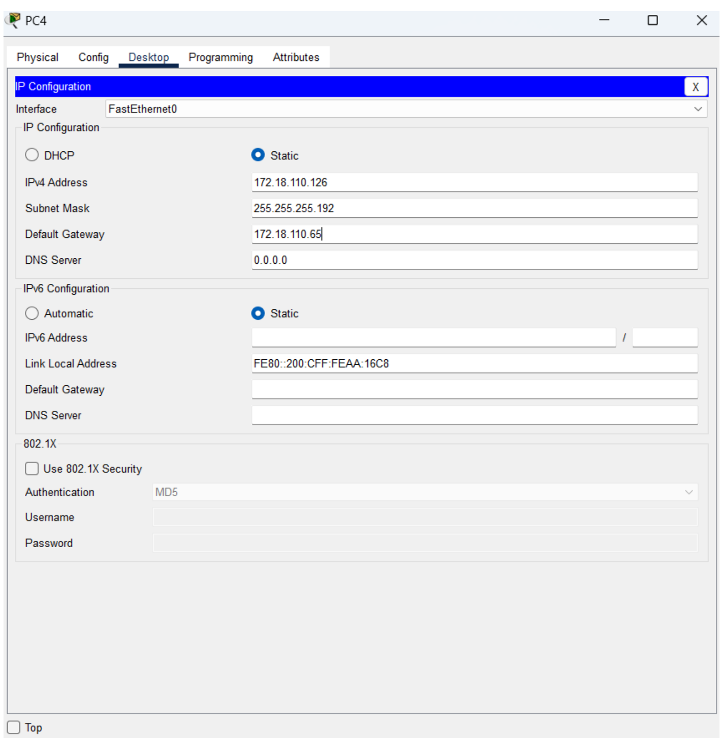
PC2


PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.18.110.62
Subnet Mask
255.255.255.224
Default Gateway
172.18.110.33
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2E0:8FFF:FED8:CCD1
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

PC3


PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.18.110.126
Subnet Mask
255.255.255.192
Default Gateway
172.18.110.65
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::204:9AFF:FE85:832
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

PC4


PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.18.110.126
Subnet Mask
255.255.255.192
Default Gateway
172.18.110.65
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::200:CFF:FEAA:16C8
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

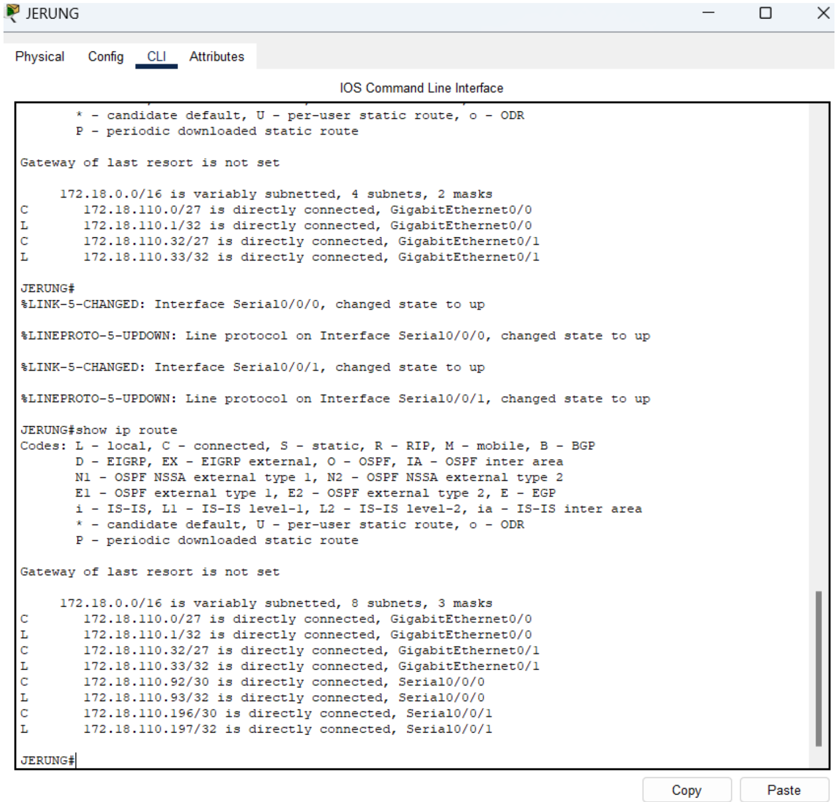
Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.
    a. CLI
        i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

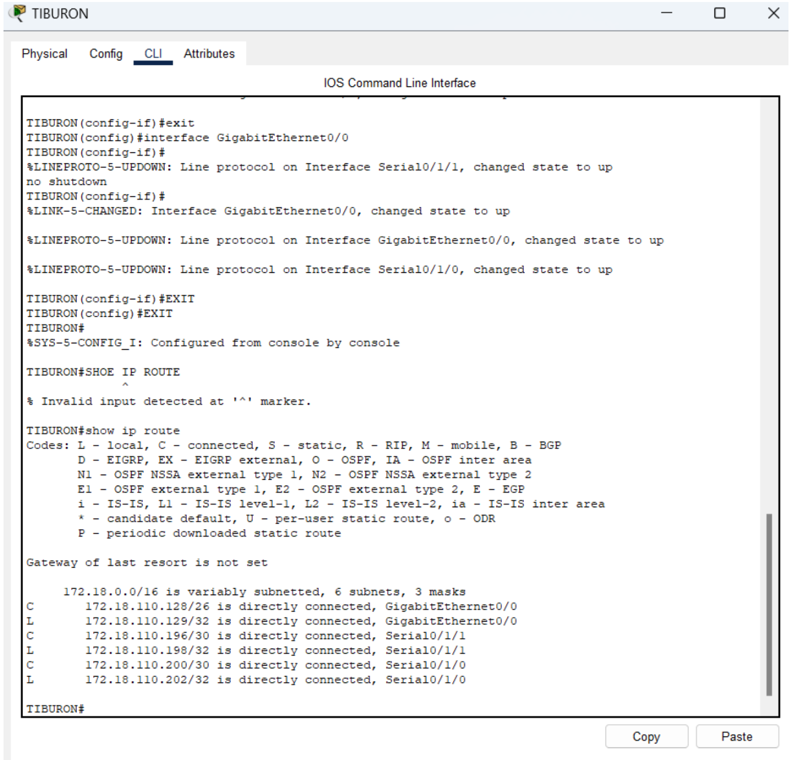
Figure A

JERUNG:



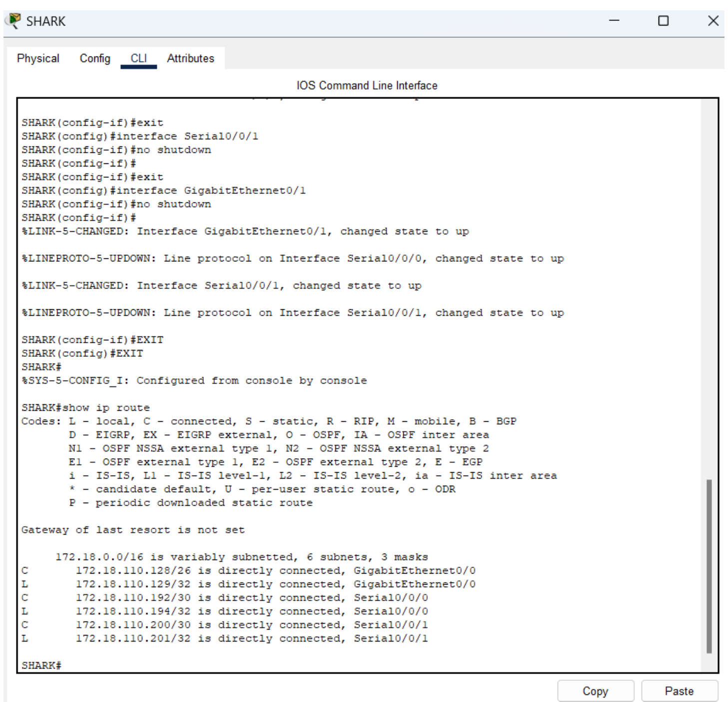

```
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 4 subnets, 2 masks
C       172.18.110.0/27 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
C       172.18.110.32/27 is directly connected, GigabitEthernet0/1
L       172.18.110.33/32 is directly connected, GigabitEthernet0/1

JERUNG#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up

JERUNG#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C       172.18.110.0/27 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
C       172.18.110.32/27 is directly connected, GigabitEthernet0/1
L       172.18.110.33/32 is directly connected, GigabitEthernet0/1
C       172.18.110.92/30 is directly connected, Serial0/0/0
L       172.18.110.93/32 is directly connected, Serial0/0/0
C       172.18.110.196/30 is directly connected, Serial0/0/1
L       172.18.110.197/32 is directly connected, Serial0/0/1

JERUNG#
```

TIBURON:




```
TIBURON(config-if)#exit
TIBURON(config)#interface GigabitEthernet0/0
TIBURON(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
no shutdown
TIBURON(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

TIBURON(config-if)#EXIT
TIBURON(config)#EXIT
TIBURON#
%SYS-5-CONFIG_I: Configured from console by console

TIBURON#SHOE IP ROUTE
             ^
% Invalid input detected at '^' marker.

TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.128/26 is directly connected, GigabitEthernet0/0
L       172.18.110.129/32 is directly connected, GigabitEthernet0/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0

TIBURON#
```

SHARK:




```
SHARK(config-if)#exit
SHARK(config)#interface Serial0/0/1
SHARK(config-if)#no shutdown
SHARK(config-if)#
SHARK(config-if)#exit
SHARK(config)#interface GigabitEthernet0/1
SHARK(config-if)#no shutdown
SHARK(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up

SHARK(config-if)#EXIT
SHARK(config)#EXIT
SHARK#
%SYS-5-CONFIG_I: Configured from console by console

SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.128/26 is directly connected, GigabitEthernet0/0
L       172.18.110.129/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1

SHARK#
```

b. PT tool

i. Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

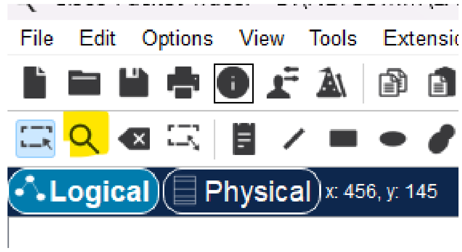


Figure B

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

JERUNG:

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
C	172.18.110.92/30	Serial0/0/0	---	0/0
L	172.18.110.93/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0

TRIBURON:

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

SHARK:

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

2. Try to ping PC2 from PC1, paste the results here.

PC1

Physical Config Desktop Programming Attributes

Command Prompt X

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172,18,110,62
Ping request could not find host 172,18,110,62. Please check the name and try again.
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

3. Try to ping PC4 from PC1, paste the results here.

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Request timed out.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

4. Explain the reason(s) behind the results.

   PC1 ping PC2 successfully because both devices are in subnets directly connected to JERUNG, allowing routing to occur correctly.

   PC1 failed to ping PC4 because they are in different subnets and there is no Configurations in SHARK that enable devices in different subnets to communicate across different router.

5. What needs to be done to ensure all PCs can ping each other successfully?

- Verify physical connectivity and IP addressing on all devices,
- Implement Routing Information Protocol on all routers (JERUNG, SHARK, TIBURON) to enable dynamic route sharing across subnets.
- Add static routes where necessary to fill gaps in the routing table.

Task 2 – Routing Configuration

1. Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.161/32	GigabitEthernet0/0	---	0/0

Figure D

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
C	172.18.110.92/30	Serial0/0/0	---	0/0
L	172.18.110.93/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0

2. In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```
JERUNG#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#
```

Figure E

a. What can you say about the addresses used in the 'network' instructions in Figure E?

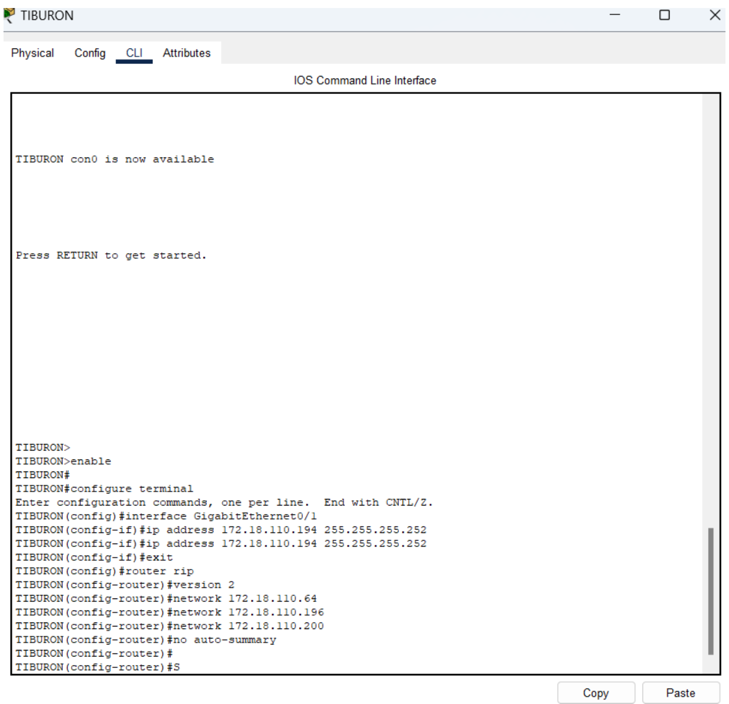
```
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#
JERUNG(config-router)#S
```

The addresses are the network addresses for different ranges of IPs. This enables devices in different subnet and router to communicate between each other.

3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```
TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200
```

TIBURON



4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.92/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
C	172.18.110.92/30	Serial0/0/0	---	0/0
L	172.18.110.93/32	Serial0/0/0	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.198	120/1
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0
R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1

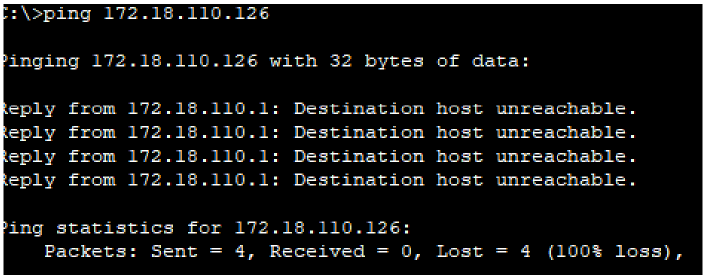
a. What are the changes seen in TIBURON?
Some networks with type R exist in the routing table.

b. What are the Networks with type R in TIBURON and JERUNG?
TIBURON: 172.18.110.0/27, 172.18.110.32/27 and 172.18.110.192/30
JERUNG: 172.18.110.128/26 and 172.18.110.200/30

c. Ping PC3 from PC1. Was it successful?
PC1 ping PC3 successfully.

```
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.190: bytes=32 time=16ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
Reply from 172.18.110.190: bytes=32 time=4ms TTL=126
Reply from 172.18.110.190: bytes=32 time=6ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 16ms, Average = 6ms

C:\>
```

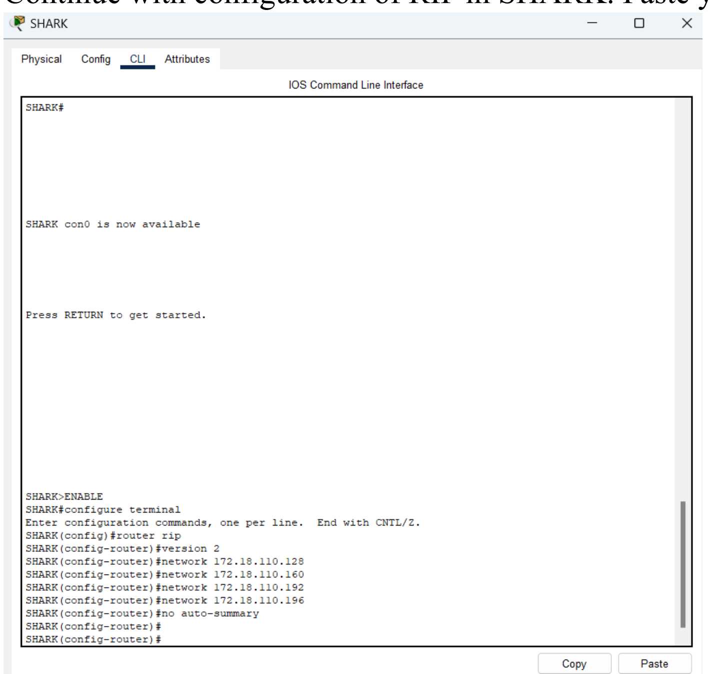
d. Ping PC4 from PC1. Was it successful?
PC1 failed to ping PC4



e. Explain the reasons for your answer.
   PC1 successfully pinged PC3 because both PCs belong to subnets that are directly connected to the JERUNG router. This allows JERUNG to route traffic between its interfaces.

   However, PC1 failed to ping PC4 because they are in different subnets that are not connected to the same router. While PC1 is connected to JERUNG, PC4 is connected to SHARK. Additionally, the SHARK router lacks a RIP configuration, which prevents proper routing between the networks.

f. Continue with configuration of RIP in SHARK. Paste your configurations here.

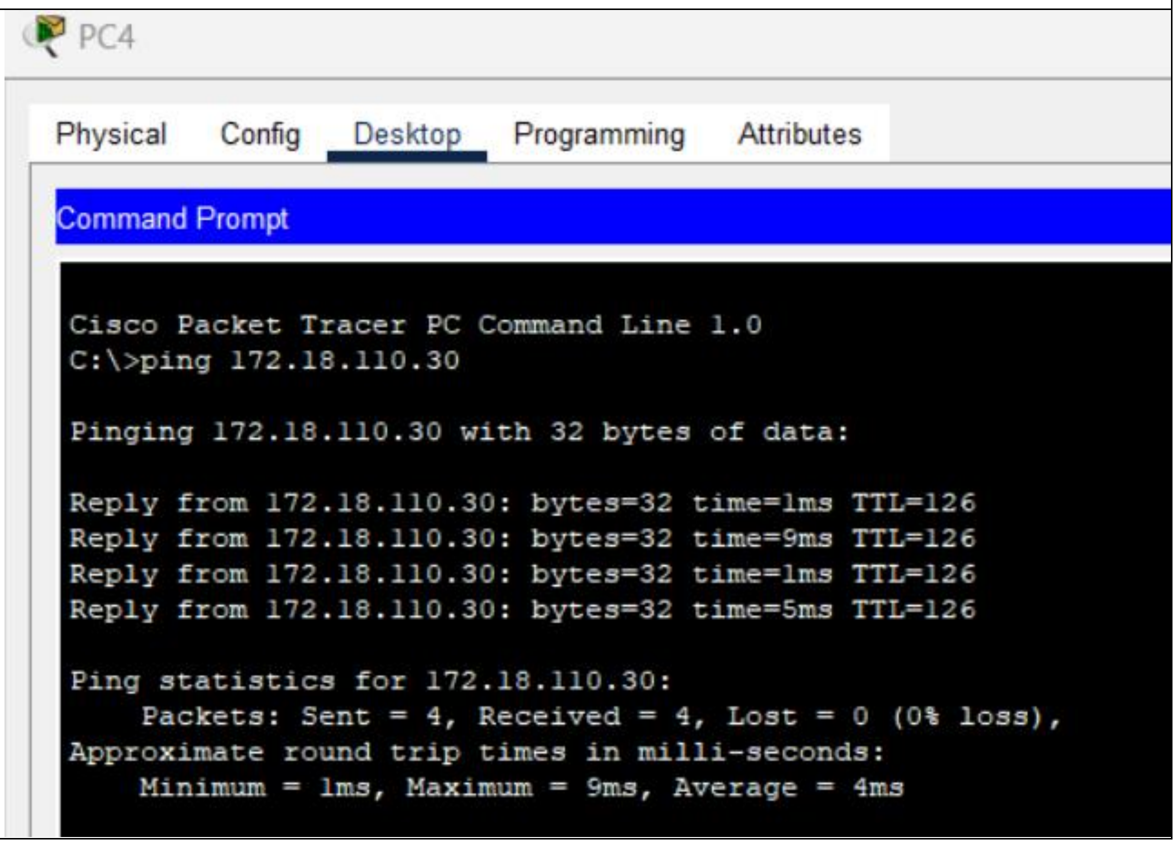


g. Open router SHARK’s routing table and paste here.

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/1	172.18.110.202	120/2
R	172.18.110.32/27	Serial0/0/1	172.18.110.202	120/2
R	172.18.110.92/30	Serial0/0/1	172.18.110.202	120/2
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

PC	Ping result
1	

2

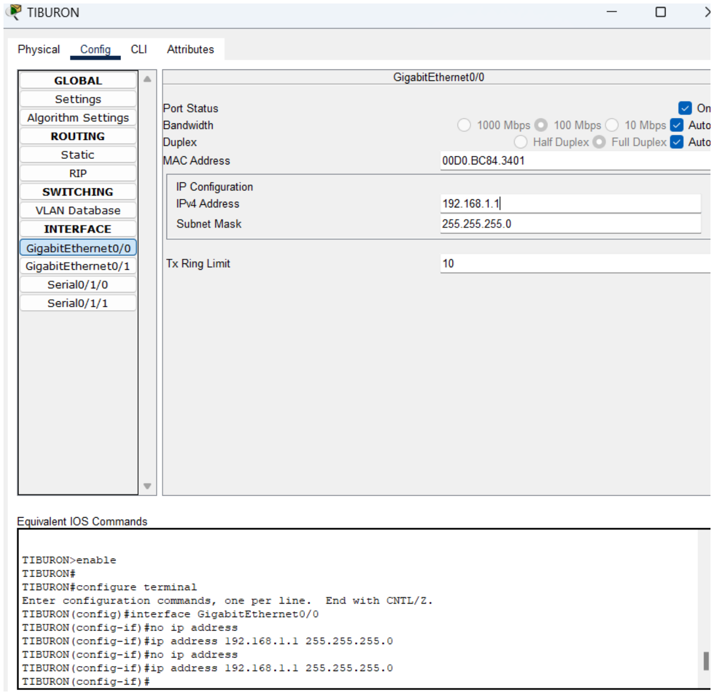
```
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.190: bytes=32 time=18ms TTL=126
Reply from 172.18.110.190: bytes=32 time=7ms TTL=126
Reply from 172.18.110.190: bytes=32 time=8ms TTL=126
Reply from 172.18.110.190: bytes=32 time=5ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 18ms, Average = 9ms
```

3

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=10ms TTL=126
Reply from 172.18.110.62: bytes=32 time=11ms TTL=126
Reply from 172.18.110.62: bytes=32 time=9ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 7ms
```

Task 3 – Routing Update

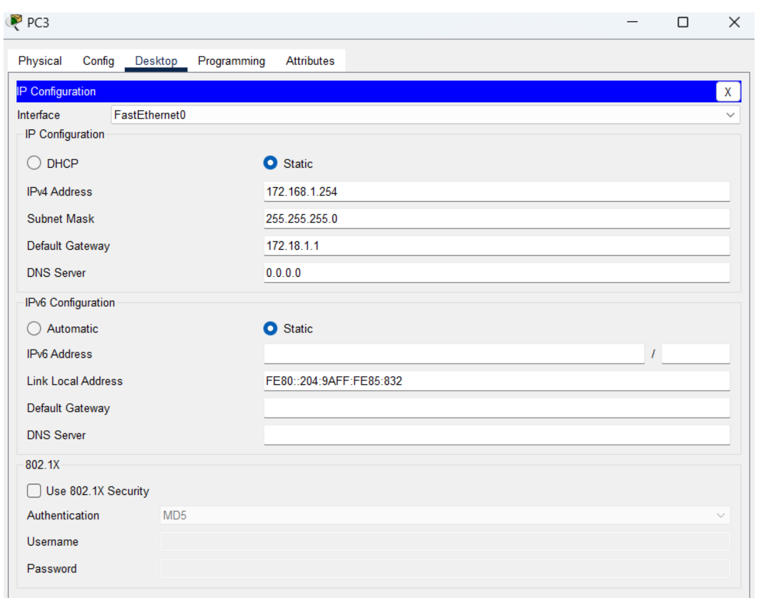
1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.



2.

   a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: 192.168.1.0/24

   b. As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?



PC3	Info
IP address	192.168.1.254
Subnet Mask	255.255.255.0
Gateway Address	192.168.1.1

c. After this change, can PC4 and PC1 ping PC3?

PC4 and PC1 cannot ping PC3.

PC4:

```
C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 172.18.110.65: Destination host unreachable.
Reply from 172.18.110.65: Destination host unreachable.
Request timed out.
Reply from 172.18.110.65: Destination host unreachable.

Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

PC1:

```
C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

d. Copy and paste the routing tables for both SHARK and TIBURON here.
SHARK:

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/1	172.18.110.202	120/2
R	172.18.110.32/27	Serial0/0/1	172.18.110.202	120/2
R	172.18.110.92/30	Serial0/0/1	172.18.110.202	120/2
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

TIBURON:

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.92/30	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.128/26	Serial0/1/0	172.18.110.201	120/1
R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0
C	192.168.1.0/24	GigabitEthernet0/0	---	0/0
L	192.168.1.1/32	GigabitEthernet0/0	---	0/0

e. Referring to the routing table, explain your findings.

SHARK's routing table is missing the 192.168.1.0/24 subnet, whereas TIBURON's routing table include it. As a result, PC4 and PC1 are unable to successfully ping PC3. To resolve this issue, both JERUNG and SHARK must include the 192.168.1.0/24 subnet in their routing tables, ensuring that all devices can communicate successfully across different subnets.

f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?

Add static routes for new TIBURON subnet (192.168.1.0/24) on JERUNG and SHARK.

JERUNG:

```
JERUNG>enable
JERUNG#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
JERUNG(config)#ip route 192.168.1.0 255.255.255.0 172.18.110.198
```

TIBURON:

```
TIBURON(config)#ip route 172.18.110.0 255.255.255.224 172.18.110.197
TIBURON(config)#ip route 172.18.110.64 255.255.255.192 172.18.110.201
```

SHARK:

```
SHARK(config)#ip route 192.168.1.0 255.255.255.0 172.18.110.193
```

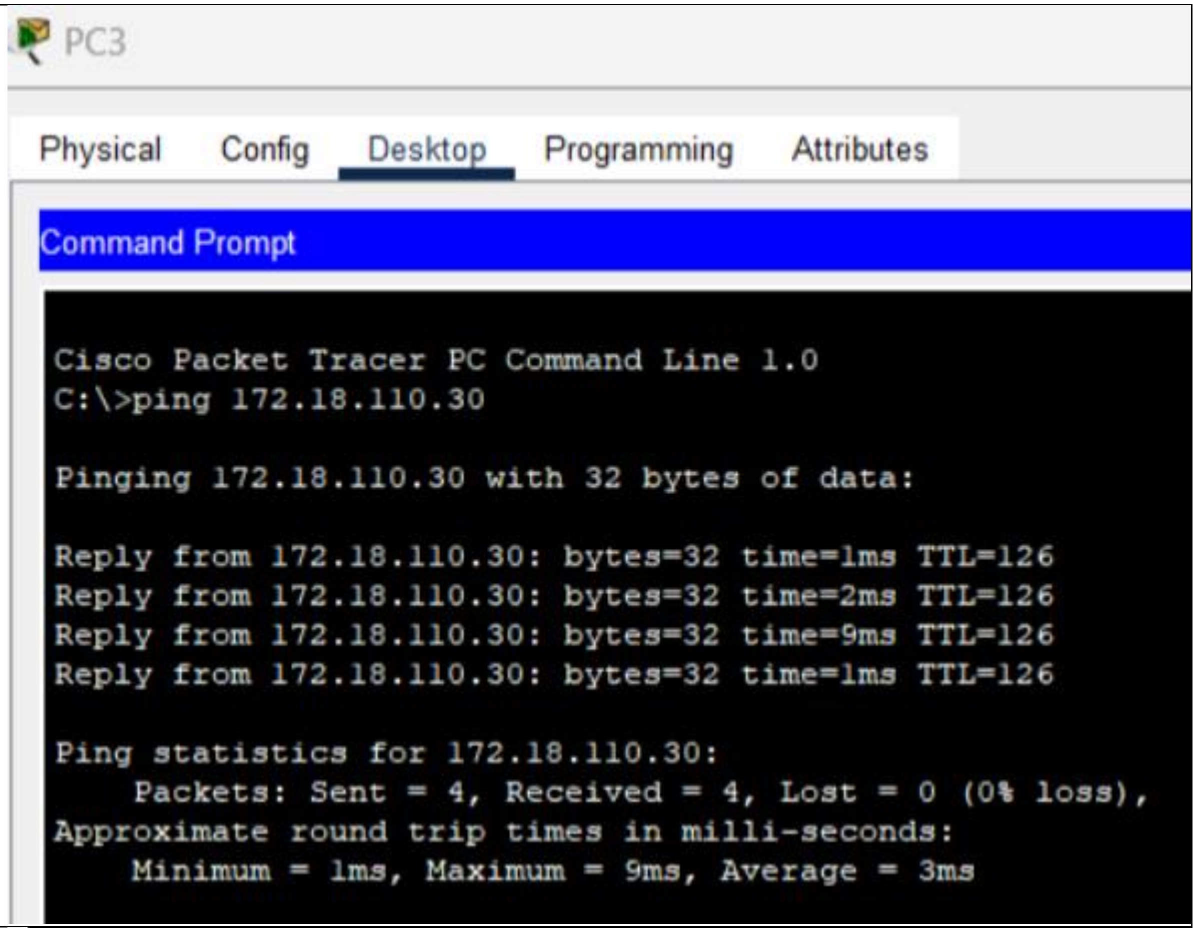
g. Show your configurations in TIBURON to ensure end-to-end connectivity.

```
TIBURON(config)#ip route 192.168.1.0 255.255.255.0 172.18.110.198
%Invalid next hop address (it's this router)
TIBURON(config)#ip route 192.168.1.0 255.255.255.0 172.18.110.201
TIBURON(config)#ip route 172.18.110.0 255.255.255.224 172.18.110.197
TIBURON(config)#ip route 172.18.110.64 255.255.255.192 172.18.110.201
TIBURON(config)#
```

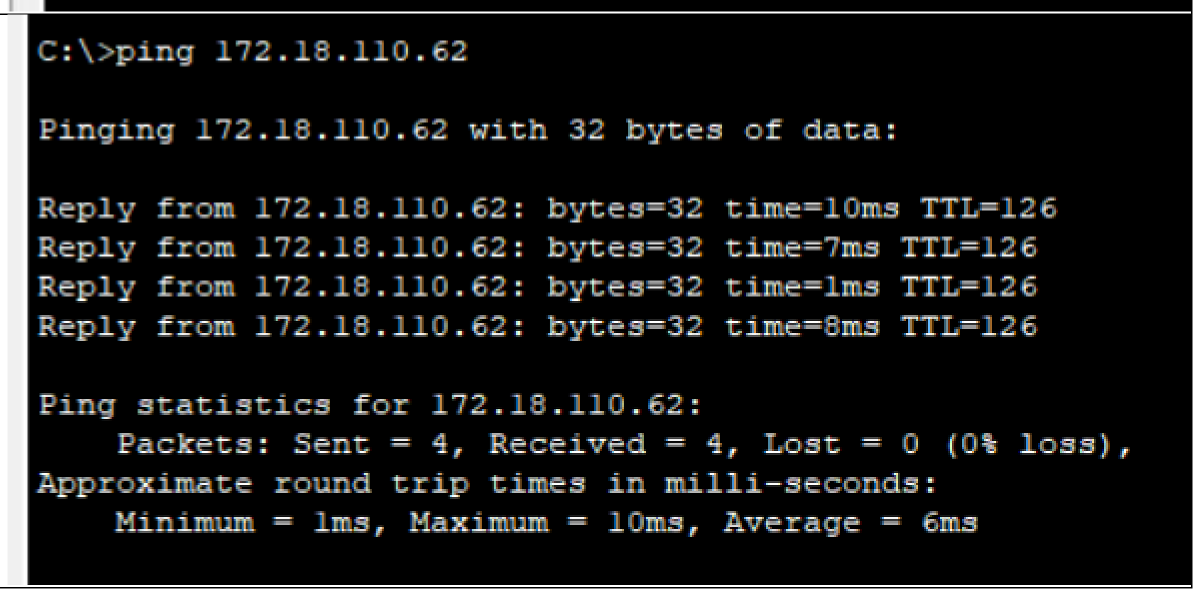
h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

PC	Ping result

1


PC3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.18.110.30
Pinging 172.18.110.30 with 32 bytes of data:
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126
Reply from 172.18.110.30: bytes=32 time=2ms TTL=126
Reply from 172.18.110.30: bytes=32 time=9ms TTL=126
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126
Ping statistics for 172.18.110.30:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 9ms, Average = 3ms

2


C:\>ping 172.18.110.62
Pinging 172.18.110.62 with 32 bytes of data:
Reply from 172.18.110.62: bytes=32 time=10ms TTL=126
Reply from 172.18.110.62: bytes=32 time=7ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126
Reply from 172.18.110.62: bytes=32 time=8ms TTL=126
Ping statistics for 172.18.110.62:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 10ms, Average = 6ms

4	C:\>ping 172.18.110.126
	
	Pinging 172.18.110.126 with 32 bytes of data:
	
	Reply from 172.18.110.126: bytes=32 time=11ms TTL=125
	Reply from 172.18.110.126: bytes=32 time=10ms TTL=125
	Reply from 172.18.110.126: bytes=32 time=17ms TTL=125
	Reply from 172.18.110.126: bytes=32 time=9ms TTL=125
	
	Ping statistics for 172.18.110.126:
	    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
	Approximate round trip times in milli-seconds:
	    Minimum = 9ms, Maximum = 17ms, Average = 11ms

REFLECTION

What have you learned in this task?

This lab helped me learn how to use Cisco Packet Tracer for subnetting, routing, and network configuration. It showed the importance of proper IP addressing and router setup for communication across subnets. I also gained experience configuring RIP protocols, adding static routes to ensure connectivity and testing configurations using the “ping” and “show ip route” commands.